

WorldEpisode: Diagnosing and Correcting Hidden World–Episode Failures in Robot-Learning Datasets

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Abstract

Robot-learning pipelines can fail while every file still looks valid: a controller may interpret degrees as radians, a replay may treat a command timestamp as an actuation timestamp, or a random train/test split may leak the same reconstructed room into both sides of an evaluation. We show that these are not bookkeeping annoyances but measurable threats to robot-learning validity. On a public ArmnetBench LeRobot release, a standard random episode split leaks every tested world lineage and gives a Torch behavioral-cloning baseline 0.850 offline imitation success; enforcing a scene-disjoint split drops leakage to zero and success to 0.000. On a real SO-101 LeRobot trajectory, ignoring the inferred four-frame, 133 ms control delay gives 4.732 degree validation joint RMSE, while timestamp-aware replay reduces it to 1.862 degrees; a tested MuJoCo replay drops from 3.425 to 1.563 degrees. We introduce *WorldEpisode*, a storage-neutral sidecar contract that prevents these silent failures by binding each episode to an immutable world revision through persistent identity, explicit space/time, representation roles, interaction events, and provenance. WorldEpisode does not replace LeRobotDataset, Rerun, NCore, MCAP, OpenUSD, or glTF Gaussian splats; it specifies the invariants those systems must preserve for reproducible conversion, replay, and evaluation. The reference implementation includes an active public LeRobotDataset-to-WorldEpisode-to-LeRobotDataset converter, executable binding artifacts, a semantic validator, and independent fixtures. Across seven bindings, native containers preserve only 17–39% of the contract, while WorldEpisode sidecars recover the complete contract for dataset/log/world bindings. The validator detects all 14 injected fault classes and all hand-authored independent fixture failures. The central result is simple: bad data plumbing can make good robots look better than they are, and explicit world–episode semantics make those failures detectable before deployment.

1 Introduction

Imagine training a behavioral-cloning policy for three weeks on a public robot dataset. The loss curves are smooth, the simulator reports an 85% success rate, and every Parquet, video, and scene file passes its local loader. Then the policy is deployed on the real arm and immediately misses the object. The failure is not a new neural architecture problem. The dataset silently encoded joint values in a unit the controller did not declare, the replay script treated command timestamps as motor-effective timestamps, and the random split let the policy see the same reconstructed room in both training and evaluation. Nothing was corrupt, but the experiment was scientifically broken.

This is the failure mode WorldEpisode targets. Modern robot-learning pipelines no longer move only arrays from disk to a policy. Recorded episodes become training datasets; reconstructed rooms become simulator assets; Gaussian splats, meshes, collision proxies, and logs move through different tools; and counterfactual worlds become augmentation. A camera transform with the

wrong direction, a millisecond command delay, or a train/test split that shares the same world lineage can change the conclusion without changing file validity.

Existing formats solve important pieces of this chain. The LeRobot library vertically integrates robot learning from motor interfaces through dataset collection, storage, streaming, policies, and deployment [3]. The current LeRobotDataset v3 layout stores high-frequency states, actions, and timestamps in Parquet, visual streams in MP4 shards, and episode indexing in metadata rather than filenames [4]. Rerun provides a physical-data layer and storage target for multimodal recordings. NCore documents explicit pose graphs and sensor conventions. OpenUSD, SimReady, glTF, and simulator-native formats describe composed worlds and assets. Gaussian splats are also moving into standard asset containers through glTF’s `KHR_gaussian_splatting` extension [9, 10]. Recent real-to-sim systems such as RoboSnap and DROID-Sim show that Gaussian-splat environments can support trajectory replay, synthetic data generation, and policy evaluation at dataset scale [8].

The missing piece is not another monolithic file. A LeRobot table can store actions without knowing which immutable world revision they came from; a USD scene can store assets without knowing the policy action semantics that generated a trajectory; a ROS or MCAP log can store messages without knowing whether evaluation splits leak the same physical room. WorldEpisode fills that gap as a sidecar contract: it binds each episode to the world in which it happened and records the assumptions required to convert, replay, and evaluate it.

WorldEpisode is:

- **storage-neutral**: serializable through LeRobotDataset v3, Rerun, NCore, MCAP, or a reference package;
- **representation-neutral**: compatible with splats, meshes, NeRFs, point clouds, and future assets;
- **runtime-neutral**: transferable to Isaac Sim, MuJoCo, SAPIEN, Genesis, and other simulators;
- **loss-explicit**: conversions may be lossy, but never silently lossy.

Figure 1 summarizes the intended relationship: WorldEpisode sits between existing dataset/log containers and world/simulator assets as a semantic contract, not as a replacement storage engine.

The paper’s core claim is therefore not a new monolithic file format. It is an executable interchange profile for world–episode semantics: identity, frames, clocks, action contracts, representation roles, events, provenance, uncertainty, and lineage-safe evaluation splits. The current artifact includes a JSON Schema, semantic validator, round-trip binding artifacts, an active public LeRobotDataset v3 conversion experiment, hand-authored independent fixtures, and controlled experiments for replay drift, leakage, and counterfactual augmentation.

2 Problem Statement and Scope

WorldEpisode starts from a practical question: what must be known before an episode can be safely converted, replayed, augmented, or used for evaluation? A valid package must answer five questions.

Which world did this happen in? Each episode references one immutable, content-addressed world revision plus an ordered list of episode-specific deltas. Changing geometry, metric scale, calibration, collision proxies, physical parameters, entity decomposition, object identity, or task-relevant fixtures creates a new world revision. This prevents two datasets from silently using the same filename for different physical worlds, or different filenames for the same world.

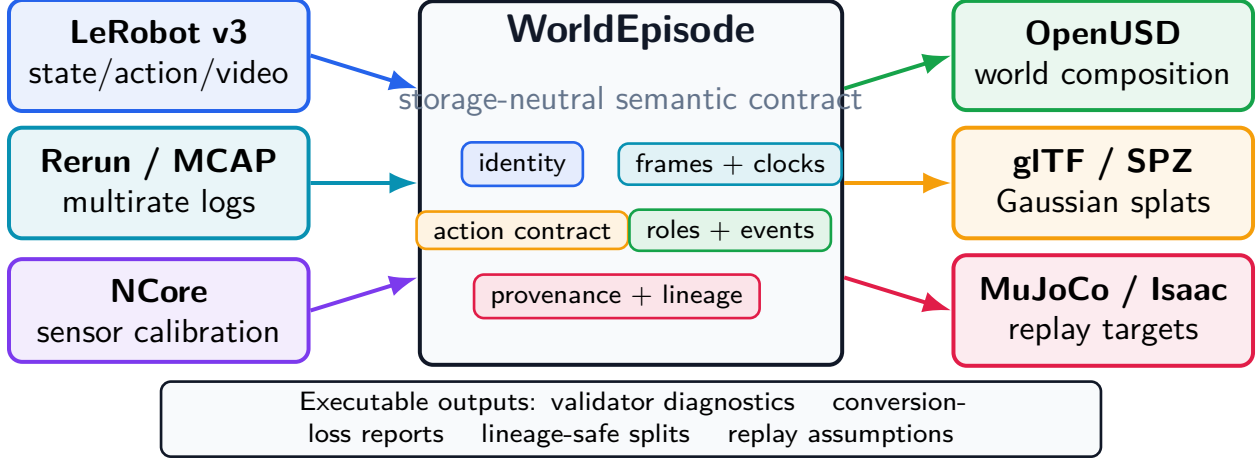


Figure 1: WorldEpisode is the semantic bridge between robot-learning containers, reconstructed world assets, and simulator targets. It preserves the world-episode contract without replacing the storage formats on either side.

What did the robot see and touch? Objects, robot links, sensors, fixtures, support surfaces, semantic regions, and generated assets must have persistent identifiers. A mug in a segmentation mask, a Gaussian group, a collision mesh, a simulator actor, a language annotation, and a contact event should resolve to the same entity when they refer to the same physical object.

Where and when did it happen? Spatial values declare their frames, units, transform direction, validity interval, and calibration revision. Streams declare clock domains, timestamps, synchronization mappings, and timing error where available. Actions declare command time and effective motor time rather than assuming those are the same instant.

What was preserved or lost during conversion? WorldEpisode reuses existing standards rather than replacing them. USD [1] carries composed simulation worlds, glTF carries Gaussian appearance assets [9], NCore carries capture and pose-graph conventions [6], Rerun carries physical-data recordings with column chunks [7], and ROS/MCAP carries logs. A converter may externalize or approximate fields, but it must emit a machine-readable loss report instead of silently discarding semantics.

Can the evaluation split leak? Episode-level random splits are not enough when multiple episodes share a reconstructed room, physical object, source capture, calibration, or generated asset lineage. A WorldEpisode split manifest can enforce disjointness by world lineage, physical entity, source capture, reconstruction run, or generated asset family.

Version 1 is intentionally narrow: rigid tabletop manipulation with fixed-base single- or dual-arm robots. Humanoids, locomotion, deformables, fluids, and multi-agent worlds are later profiles. This keeps conformance executable rather than aspirational.

3 The Five-Graph Model

Figure 2 shows the five coupled graphs that make an episode replayable and auditable rather than just stored. Each graph answers a question that appears during debugging: what object is this,

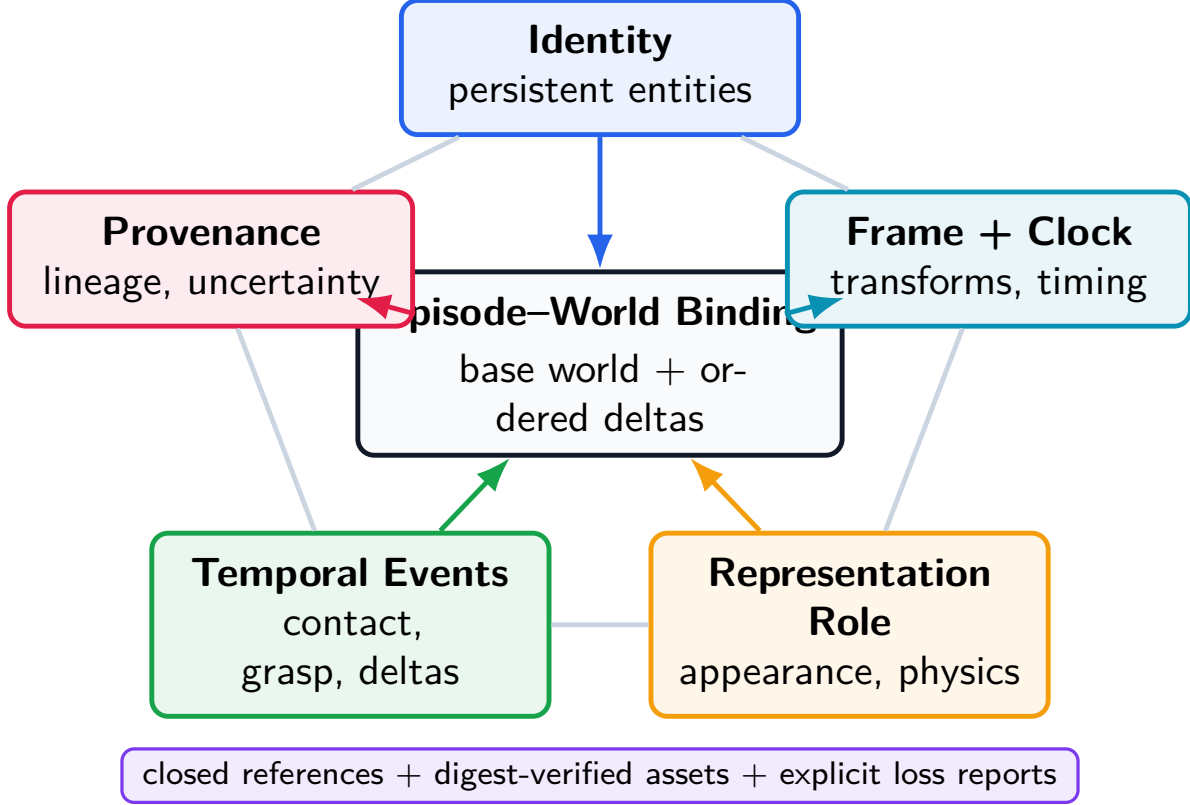


Figure 2: The five-graph model. WorldEpisode makes identity, frames/clocks, representation roles, temporal events, and provenance mutually referential so conversion and replay errors become detectable instead of hidden.

where and when is it, what is the asset safe to do, what physically happened, and where did this data come from?

3.1 What object am I looking at?

Every physical or logical entity receives a persistent identifier: robot, link, gripper, sensor, rigid object, articulated object, fixture, support surface, region, human, light, or semantic group. The same `entity_id` must be used by segmentation masks, bounding boxes, Gaussian groups, render meshes, collision bodies, simulator actors, language references, contact events, object trajectories, grasp events, and attachment events. The real-world fix is simple: if a language command says “pick up the mug,” the perception mask, the simulator object, and the collision proxy must identify the same mug.

3.2 Where is it, and exactly when?

Every spatial value declares source frame, target frame, units, handedness, up axis, transform direction, quaternion convention, validity interval, calibration revision, and uncertainty where available. Every stream declares a clock domain, device timestamp, host-receive timestamp where available, exposure interval for cameras, effective command time for actions, synchronization mapping, and estimated offset/drift/error. NCore’s explicit pose-graph convention demonstrates the kind of rigor WorldEpisode requires across the whole episode [6]. This graph is what turns a

suspicious replay from “the policy is bad” into a diagnosable offset, transform direction, or unit error.

3.3 What is this asset safe to be used for?

One physical entity can have many representations: Gaussian splat appearance, textured render mesh, metric surface mesh, depth point cloud, collision proxy, mass/inertia record, semantic affordance, or learned embedding. Each representation declares its role and an asset descriptor: URI, media type, SHA-256 digest, optional mirrors, optional embedded payload metadata, coordinate frame, units, validity interval, derivation, uncertainty, and license. The URI may be relative, HTTP(S), Hugging Face, object storage, OCI, IPFS, or another registered resolver. Portability comes from deterministic resolution and digest verification, not from requiring every asset to live in one folder. The role field is essential: a splat may be valid for appearance but invalid for collision, while a simplified proxy may be valid for collision but unsuitable for visual training.

3.4 Did the robot actually grab the object?

Episodes do not store only poses. They record interaction events: contact begin/end, grasp establish/release, attachment create/remove, support, containment, articulation state changes, spawn/despawn, topology changes, simulator reset, intervention, subgoal completion, and failure detection. Each event references entities, time, confidence, source, and optionally evidence. This turns raw coordinate streams into physical milestones such as “contact established,” “object dropped,” or “intervention occurred.”

3.5 Where did this data come from?

Every derived asset identifies source episodes, source sensors and intervals, reconstruction algorithm, software version, model/checkpoint hash, configuration, calibration revision, manual edits, source and output hashes, creator, license, and quality metrics. WorldEpisode maps to existing provenance and dataset metadata vocabularies rather than inventing an incompatible publication vocabulary. This graph lets a split avoid putting the same reconstructed room, capture session, or generated asset family on both sides of evaluation.

4 Action Semantics, Revisions, and Loss Reports

4.1 Complete Action Semantics

An action vector is not portable without its operational contract. Each action channel must declare actuator, control mode, parameterization, reference frame, units, absolute/delta/velocity semantics, limits, command rate, command timestamp, effective timestamp, latency model, interpolation, normalization, gripper semantics, and missing-value policy. A vector such as $[0.01, 0, 0, 0, 0, 0, 1]$ is otherwise ambiguous: it could be a tool-frame delta, base-frame delta, velocity command, normalized policy output, or delayed target increment.

4.2 Immutable World Revisions

Each episode references one base world revision plus ordered episode deltas. A new revision is required when world geometry, metric scale, calibration, collision proxies, physical parameters, entity

decomposition, object identity, or task-relevant fixtures change. Content hashes allow two datasets to prove that they refer to the same world without relying on filenames.

4.3 Loss-Aware Conversion

WorldEpisode supports many bindings, but it does not promise impossible universal losslessness. Every converter emits a machine-readable report listing preserved, externalized, approximated, discarded, and warning fields. The standard permits lossy conversion; it forbids silent lossy conversion.

4.4 Implementation Artifact

We implement the contract as four executable components: a JSON Schema for structural validity, a machine-readable conformance requirement file, a semantic validator, and a round-trip experiment runner. The validator maps each diagnostic to a stable requirement identifier such as `TIME.001`, `FRAME.002`, or `ASSET.002`. The experiment runner takes a WorldEpisode package, exports binding-native artifacts plus sidecars, reimports them, compares a semantic projection, injects controlled faults, and writes reproducible result artifacts. URDF Studio remains the initial implementation root, but the contract is represented independently in the public schema and validator.

The repository also includes an active LeRobotDataset v3 converter. It downloads bounded Parquet and metadata shards from the pinned public `lerobot/svla_so101_pickplace` SO-101 dataset, constructs a schema-valid WorldEpisode manifest and sidecar, exports a small LeRobot-Dataset v3 package, reads that package back, and asserts equality of source-native tensors and timestamp records. Source fields that LeRobot does not declare, such as camera extrinsics and controller latency, are not guessed; they are emitted in the conversion report as explicitly source-absent. The split-leakage tool similarly downloads bounded ArmnetBench LeRobot shards, writes `world_lineage` and split-manifest artifacts, and trains an executable BC baseline over the real LeRobot tensors. The control-replay tool reads the exported LeRobot trajectory, estimates the effective command delay from action/state alignment, writes the WorldEpisode action contract, executes the tested MuJoCo adapter, and emits an Isaac adapter mapping that is ready for an Isaac environment but explicitly marked untested in the committed report.

5 Controlled Experiments

The experiments are written as failure investigations rather than as isolated unit tests. Each one starts from a failure mode that can make a robot-learning result look better, cleaner, or more reproducible than it really is.

Case 1: the split that faked success. The most dangerous failure is not a crash; it is a benchmark that looks successful for the wrong reason. We audit ArmnetBench v0.1’s single-arm SO-101 LeRobot release [2] by deriving eight WorldEpisode-style `world_lineage` hashes, one for each task-scene and camera-layout group. A standard random episode split trains on 320 episodes, tests on 80, and leaks all test scene lineages into training. The same Torch MLP state-action behavioral-cloning baseline then reaches 0.850 offline imitation success under a fixed normalized-RMSE tolerance of 0.25.

When the split is made scene-disjoint, the apparent success disappears. The held-out eye-drop scene lineages have zero overlap with training, the test set contains 100 episodes, and offline BC success drops to 0.000. Mean episode normalized RMSE rises from 0.183 to 0.363, a $1.98\times$ increase.

This is an offline action-imitation benchmark, not a physical rollout claim, but it exposes the scientific risk: a conventional split can reward memorizing a background world rather than learning a transferable manipulation behavior.

Figure 3 summarizes the three experiment families whose outcomes change when world-episode semantics are made explicit.

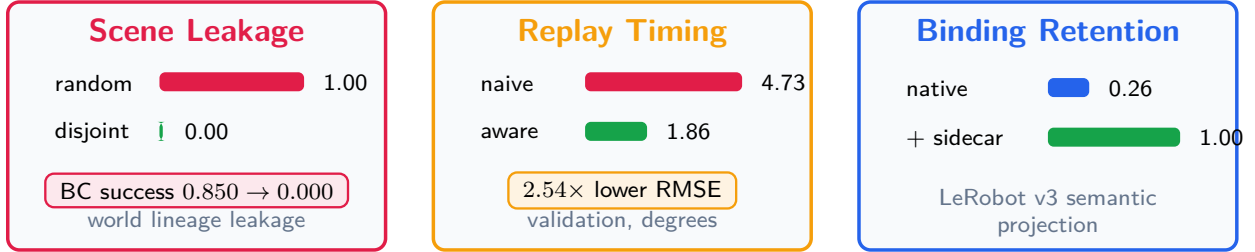


Figure 3: Three silent failures become measurable when world-episode semantics are explicit: scene leakage, replay timing drift, and native binding loss.

Case 2: the converter that refuses to invent missing physics. To test an actual LeRobotDataset rather than only a capability model, we implement LeRobotDataset v3 → WorldEpisode → LeRobotDataset v3 on episode 0 of the pinned public SO-101 dataset `lerobot/svla_so101_pickplace` [5]. The converter downloads the real Parquet metadata and data shards, builds a schema-valid WorldEpisode manifest and sidecar, exports a compact LeRobotDataset v3 package, reads it back, and asserts exact equality.

The round trip preserves 303 rows of six-dimensional action tensors, 303 state rows, 303 sample timestamps, and two camera video timestamp ranges with maximum absolute error 0.0. It also preserves three physical-frame records through the sidecar. More importantly, it does not fake missing semantics: camera extrinsics, robot/world calibration, action units, and controller latency are reported as source-absent rather than guessed. This is the difference between a convenient conversion and an auditable one.

Case 3: the replay that drifted because time was underspecified. LeRobot’s control stack can decouple policy inference from the fixed-rate low-level controller: commands may be queued, chunked, interpolated, or consumed after policy output [3]. On the public SO-101 trajectory, WorldEpisode records policy-output, enqueue, queue-consume, and motor-receive timestamp semantics, plus zero-order-hold interpolation and a constant-frame latency model.

Estimating the effective delay on the calibration prefix gives four 30 Hz frames, or 133 ms. On the validation suffix, naive command-time alignment has joint RMSE 4.732 degrees. Timestamp-aware alignment has RMSE 1.862 degrees, a 2.54× improvement. In a tested MuJoCo six-joint position-servo replay, naive command-time replay has RMSE 3.425 degrees and timestamp-aware replay has 1.563 degrees, a 2.19× improvement. The Isaac adapter is emitted as a ready mapping from the same contract, but is not tested in this environment. The failure avoided here is a common one: blaming a simulator or controller when the dataset never said when the command actually took effect.

Case 4: the validator that catches physically impossible packages. We inject 14 faults into the baseline package: missing clock domains, incomplete cross-clock mappings, unknown frames, missing transform intervals, unknown event entities, missing representation roles, digest

errors, action-contract omissions, non-content-addressed worlds, missing delta sequences, missing provenance, and missing quality records. The validator detects all 14 expected requirement failures with 1.000 recall and 0.933 precision. Two hand-authored independent fixtures also trigger the expected diagnostics. The single false-positive requirement is useful rather than harmful: removing an action reference frame triggers both `ACTION.001` and the downstream `FRAME.001` reference check.

The same suite includes a simple unit sanity check from `lerobot/svla_so101_pickplace`. The first mirrored SO-101 frame has six joint values with maximum magnitude 98.924 in degrees and 1.727 after radian conversion. Treating those degree values as radians would violate the action/state unit contract even though the tensor shape is valid.

Case 5: augmentation that uses the world rather than a side file. We evaluate a shifted-camera/object-displacement regression task. Observations-only nearest-neighbor control succeeds on 0.292 of shifted test cases. Adding unstructured 3D side information increases success to 0.342. WorldEpisode-style world-frame identity and counterfactual camera augmentation increase success to 0.592 over 120 deterministic test cases.

Binding retention. The same executable artifacts measure how much of the semantic contract is preserved by native containers alone. Each binding writes a native payload and a WorldEpisode sidecar, then the experiment reimports both and checks exact equality against a 23-field semantic projection. Table 1 shows that common containers preserve only 17–39% of the contract natively, while the sidecar recovers all fields for dataset/log/world bindings. A Gaussian asset binding alone preserves only 17% natively and 30% with limited sidecar metadata, confirming that an asset format is not an episode–world contract.

Table 1: Semantic field retention in the pilot binding model.

Binding	Native	With WorldEpisode
LeRobot v3	0.261	1.000
Rerun RRD	0.348	1.000
NCore	0.304	1.000
MCAP/ROS2	0.261	1.000
OpenUSD/SimReady	0.391	1.000
glTF Gaussian asset	0.174	0.304

Finally, we generate episodes with world/entity lineage labels and evaluate a memorization policy. A random episode split leaks every tested world/entity pair and obtains 1.000 accuracy. World-disjoint and entity-disjoint splits have zero measured leakage and reduce memorization accuracy to 0.500 and 0.562 respectively. This pilot demonstrates why split manifests need lineage constraints beyond the public ArmnetBench audit above.

6 Discussion

Why this is not only infrastructure. The experiments show three ways a result can be wrong while the files are valid: a split can leak world lineage, replay can drift because control timing is underspecified, and conversion can discard physical semantics that the target container has no native place to store. WorldEpisode is useful only because these failures are observable and testable.

The sidecar is the mechanism; the scientific claim is that explicit world–episode semantics change what a robotics result means.

Not another monolithic binary. WorldEpisode has a small semantic core and multiple bindings. LeRobotDataset v3 is the policy-training view; Rerun is the multirate physical-data view; NCore is the sensor/reconstruction capture view; MCAP/ROS is the raw log view; OpenUSD/SimReady is the composed simulation-world view; glTF or SPZ is the Gaussian asset view; and a reference directory is the conformance profile.

Lineage-safe evaluation. Episode-level random splits can leak the same reconstructed room, physical object, Gaussian asset, source video, world revision, or generated asset lineage across training and evaluation. WorldEpisode supports split constraints such as `world_lineage`, `physical_entity`, `source_capture`, `reconstruction_run`, and `generated_asset_family`.

Standards alignment. The project aligns with standards bodies rather than antagonizing them. Static 3D robot-map exchange, humanoid dataset lifecycle, OpenLABEL annotation, glTF Gaussian splats, USD composition, and NCore capture conventions are all useful neighbors. WorldEpisode is an executable profile and bridge across those layers.

Scope. The controlled experiments are deliberately scoped to expose individual failure mechanisms rather than to claim full robot-learning generality. They do not replace a large multi-lab benchmark, but they do establish executable tests for the core methodological risks: conversion loss, invalid physical semantics, replay timing drift, and lineage leakage. The next empirical step is to run the same protocol on larger multi-camera manipulation datasets and independently maintained adapters.

7 Conclusion

Robot-learning failures often arrive disguised as clean data: valid tensors, valid videos, valid scene files, and invalid assumptions. WorldEpisode makes those assumptions explicit. It binds an episode to an immutable world revision, gives persistent identity to physical entities, declares space/time/action semantics, assigns roles to assets, records events and provenance, and reports conversion loss instead of hiding it. In the reference experiments, those requirements expose an 85% random-split result that collapses to 0% under scene-disjoint evaluation, reduce real LeRobot replay drift by modeling a four-frame action delay, and catch physically incoherent packages before they become training or deployment bugs. Gaussian splats remain a high-value appearance profile, but the durable contribution is the world–episode contract around them. The next step is to scale the same protocol to larger multi-robot datasets and independently maintained bindings.

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